Vertex Cover ILP, LP Relaxation and Integrality Gap.

Review Learning Objectives

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Due Feb 18, 11:59 PM IST

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1. We wish to find the optimal vertex cover for a graph G. Solving the LP relaxation of the vertex cover problem yields an optimal solution with value 442. What can we say about the size of the optimal vertex cover of the graph G? Choose all the options that are correct. **3 / 3 points**

- [ ] Nothing can be said about the size of the optimal vertex cover from just the given information.
- [x] Using the rounding procedure on fractional solutions of the LP relaxation would yield a vertex cover of size between 442 and 884.

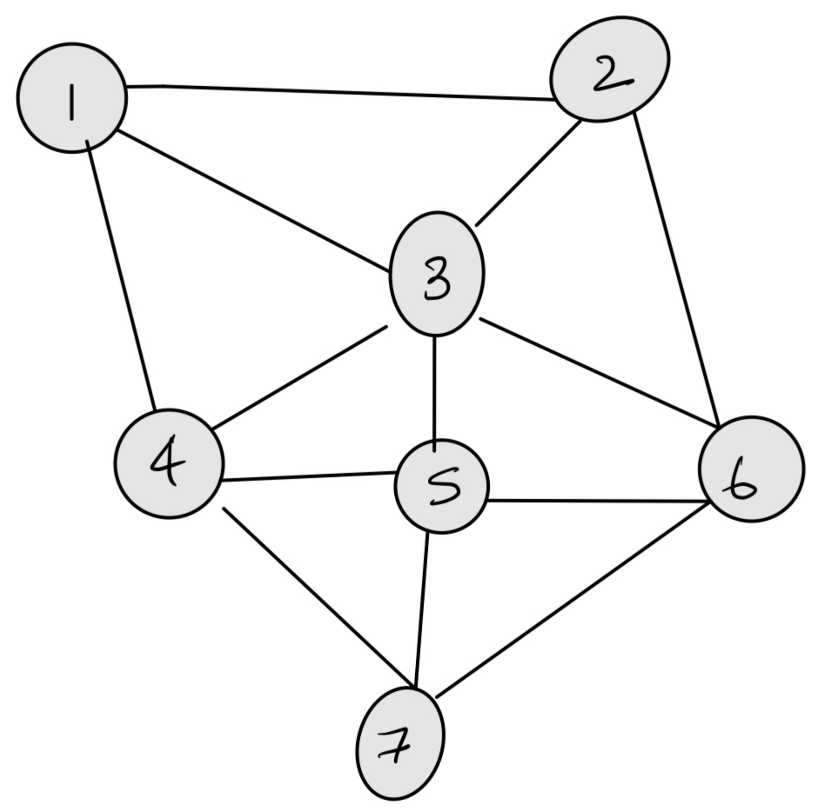
  Correct
  Correct.

- [x] The optimal vertex cover size lies in the range [442, 884].

  Correct
  Correct: the integrality gap is bounded by 2, and it is a minimization problem.

- [ ] Using the rounding procedure on fractional solutions of the LP relaxation would yield a vertex cover of size between 221 and 442.
- [ ] The optimal vertex cover size will be in the range [221, 442].

2. Calculate the optimal vertex cover for the graph below. Feel free to use the code/notes provided as part of the ungraded lab. Hopefully you have a python setup either on your computer with the pulp library installed or you are using google colab as indicated in the previous quiz on solving ILPs. **4 / 4 points**



5

Correct

3. Solve the LP relaxation for the graph in the previous problem and write down its optimal objective value below. **2 / 2 points**

3.5

Correct

4. If we rounded the LP relaxation for the problem above, what is the vertex cover we obtain? **1 / 1 point**

- ( ) The vertex cover obtained turns out to not cover two edges in the graph.
- ( ) None of the above
- ( ) The vertex cover coincides with that of the ILP
- (•) All the vertices of the graph are part of this cover.

  Correct
  Correct.